



Key Characteristics of Microservices

Building Scalable, Resilient Services



Introduction to Microservices



Microservices are small, focused units

Each service targets a specific business function



Enables scalable and maintainable systems

Easier to update, test, and deploy independently



Understanding traits is key to success

Traits guide design for resilience and efficiency



Single Responsibility Matters



Focus on One Business Capability

Design each service around a distinct business function.



Establish Clear Service Boundaries

Well-defined boundaries simplify management and scaling.



Avoid Mixing Unrelated Functions

Combining diverse responsibilities leads to complexity.



Autonomy and Independence



Services Operate Independently

Each service makes decisions without waiting for others.



Loose Coupling Boosts Resilience

Minimal dependencies mean fewer points of failure.



Graceful Handling of Failures

Autonomous services continue functioning even if others fail.



Owning Data and Communication

Service-Specific Data Ownership

Each microservice maintains its own database, ensuring clear boundaries.

No Shared Databases

Direct database sharing between services is strictly avoided to reduce coupling.

API-Driven Data Access

Services interact through APIs, not by accessing each other's databases.

Lightweight Communication Protocols

REST, messaging, and event streaming are preferred for efficient service interaction.

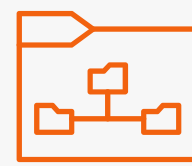


Deploy, Govern, and Isolate



Independent Service Deployment

Each service can be released or rolled back on its own.



Technology Stack Freedom

Teams select tools and frameworks best suited for their needs.



Failure Containment

Issues in one service do not disrupt others, ensuring system stability.



Observability and Automation



System Insights Through Observability

Leverage logs, metrics, and traces for deep visibility



Automated Build, Test, and Deployment

Streamline processes for reliability and speed



Minimize Manual Intervention

Automation enables scaling and rapid recovery



Checklist and Key Takeaways



Design with Clear Responsibility

Ensure each service has a focused business purpose.



Promote Autonomy and Data Ownership

Services operate independently and manage their own data.



Enable Observability and Automation

Implement logs, metrics, and automated processes for reliability.